



BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL



MINISTRY OF SCIENCE AND EDUCATION
REPUBLIC OF AZERBAIJAN

**The Ministry of Education and Science of the Azerbaijan Republic
The State Oil Company of the Azerbaijan Republic
Baku Higher Oil School**

Course Syllabus

Course Basics	
Department	Information Technology Department
Program	Information Security program
Course code / Title	IS 253 – Programming Language
Credit (ECTS)	6 ECTS
Level (Bachelor / Master)	Bachelor level
Academic year	2026/2027
Semester	Year 2, Fall semester
Group	IT25.1
Language of instruction	English
Relation to curriculum	Compulsary
Teaching methods	Lecture, Laboratory works, Home Assignments, Exams.
Workload	Total workload: 180 h = 132 h extracurricular hours + 48 h classroom Classroom hours: Lecture: 24 h (2 h /week) Laboratory work: 24 h (2 h / week) Contact hours: Examination preparation, consultation, selfstudy = 11 h/ week
Required and recommended prerequisites	- None prerequisite course is required - Mathematics (basic arithmetic operations)



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Meeting times & location	Lecture & Laboratory works – on based of Timetable Online (Microsoft Teams)		
Instructor name & title	Adjunct Instructor, Elchin Salahzada		
Contact email	elchin.salahzade@bhos.edu.az		
Office hours	Tuesday, Thursday, Saturday 4:00 - 5:30 PM		
Course description	This course provides a comprehensive introduction to programming concepts using Python, one of the most widely used and versatile programming languages in the software industry. Students will learn the fundamentals of Python syntax, data types, control structures, and functions, advanced topics such as object-oriented programming, error handling and debugging, working with libraries and modules, and file handling and data persistence.		
Course goals	By the end of this course, students will be able to: - Explain the fundamental concepts and principles of programming logic, data structures, and software design used in real-world applications. - Apply Python syntax and techniques for writing, organizing, and managing functions, classes, modules, and data across multiple program components. - Implement error handling, testing procedures, and version control mechanisms to ensure code reliability and maintainability. - Analyze program performance, algorithmic efficiency, and code structure, assessing trade-offs between readability, complexity, and scalability. - Develop and maintain complete software applications using industry-standard tools and libraries to support efficient data processing and program execution. - Interpret and evaluate problem requirements in the context of real-world applications, identifying appropriate programming approaches and solutions for specific business or engineering problems. - Work effectively in teams to design, implement, and test complete software projects from concept to deployment, demonstrating technical documentation and communication skills.		
Learning objectives			
Course schedule			
WEEK	Topic	Required Reading	Assignment



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1	Introduction to Python & Programming - Course overview - Environment setup - Programming languages - Data types	"Python Crash Course" by Eric Matthes (Chapter 1–2)	Installing Python/IDE (VS Code or PyCharm), writing and running first scripts.
2	Operators, Expressions & I/O - Operators - Type conversion - Input/Output & string formatting	"Python Crash Course" by Eric Matthes (Chapter 2)	Writing simple interactive programs (calculators, unit converters).
3	Control Structures Joins - Conditional statements - Loops	"Automate the Boring Stuff with Python" by Al Sweigart (Chapter 2)	Building decision-based and loop-driven mini programs (number guessing game, pattern printing).
4	Data Structures I: Lists & Tuples - Indexing & slicing - List methods & comprehensions - Mutability vs. immutability	"Python Crash Course" by Eric Matthes (Chapter 3–4)	List manipulation exercises, building a to-do list program.
5	Data Structures II: Dictionaries & Sets - Operations - Nested data structures	"Automate the Boring Stuff with Python" by Al Sweigart (Chapter 5)	Working with structured data (contact book, inventory tracker using dictionaries).
6	Midterm Exam		
7	Functions & Scope - Defining/calling functions - Parameters - Scope, lambda functions	"Python Crash Course" by Eric Matthes (Chapter 8)	Refactoring earlier scripts into reusable functions.
8	Object-Oriented Programming I - Classes & objects - Attributes	"Python Crash Course" by Eric Matthes (Chapter 9)	Designing simple classes (e.g., a Student or BankAccount class).



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9	Object-Oriented Programming II - Inheritance & polymorphism - Method overriding	"Fluent Python" by Luciano Ramalho (Chapter 11)	Extending base classes into subclasses; building a small class hierarchy project.
10	Error Handling, Debugging & File I/O - Debugging techniques - Reading/writing files	"Automate the Boring Stuff with Python" by Al Sweigart (Chapter 9–10)	Building a file-based data logger; handling exceptions gracefully.
11	Modules, Libraries & Data Handling - Importing modules/packages - Virtual environments - Built-in methods	"Python Crash Course" by Eric Matthes (Chapter 16)	Setting up a virtual environment, using an external library in a small script.
12	SDLC & Further improvements - Software Development Life Circle - Finalizing course materials - Preparation Exam	"Python Crash Course" by Eric Matthes (Chapter 18–19, project chapters)	Designing and building a small end-to-end program (e.g., a simple contact manager or budget tracker) using concepts from the whole course.
	Final Exam		
Course materials		Required textbooks [1] "Python Crash Course" by Eric Matthes [2] "Automate the Boring Stuff with Python" by Al Sweigart [3] "Fluent Python" by Luciano Ramalho Recommended resources Teaching materials: Presentations (Lecture Notes) on the Programming Language (on site: www.lms.bhos.edu.az) Software: Python, IDLE (Visual Studio Code or PyCharm) For class presentations and discussions, the student should utilize journal and internet materials. Moreover, the course does not limit the use of learning materials available at BHOS library.	



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Assignments details	Total number of Assignments/Lab is ten. Each lab is graded separately. At the end of the term, the average mark of the labs is calculated.		
Assessment structure	Assessment Component	Weight (%)	Description
	Final exam	40%	Final exam at the end of semester
	Midterm exam	30%	6 th –7 th week of the semester. Midterm covers the half of semester topics.
	Laboratory/Assignment	20%	During class or Home Assignment, 1 lab/Assignment per week.
	Class Activity	10%	Activity grades are given individually to students. It shows how active a student is during class.
	Total: 100%		
Grading criteria	For local students		For foreign students
	A 91-100 - Excellent B 81-90 - Very good C 71-80 - Good D 61-70 - Satisfactory F ≤ 60 - Fail		A 91-100 - Excellent B 81-90 - Very good C 71-80 - Good D 61-70 - Satisfactory E 51-60 - Pass F ≤ 50 - Fail
Study and examination requirements	Exams To be excused from the exam, the student must contact the dean and the instructor before the exam. Excuse will not be granted for social activities such as trips, cruises and sporting events (unless you are participating). The exams will all be cumulative. Most of the questions on each exam will be taken from the chapters covered since the last exam. But some will come from the earlier chapters. In general, the coverage will reflect the amount of time spent in class on different chapters.		



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	Withdrawal (pass / fail) This course strictly follows the grading policy of the Information Technology Department. Thus, a student is normally expected to achieve a total mark (pre-exam score + exam score) of at least 61 (for foreign students -51) to pass. In this case of failure, he/she will be referred or required to participate in Resit exam.
Student responsibilities	
Class participation expectations	1. Attend class regularly, arrive on time, leave only when dismissed 2. Attend class with all materials required, be prepared to listen and work 3. Be well prepared for class, read all required materials, and complete all necessary preparation 4. Be attentive in class, take notes, contribute to discussion and ask intelligent questions 5. Demonstrate professional and respectful interpersonal relationships with peers and instructor: ATTITUDE COUNTS, AND whining is unacceptable 6. Take responsibility for your actions, and your results
Attendance requirements	The students are required to attend all classes as a part of their studies, and those having legitimate reasons for absence (illness, family bereavement, etc.) are required to inform the instructor. There is a limitation for Absence –25%. After reaching this limit the student will not be allowed to Final Examination
Homework submission rules	Students are expected to submit all homework assignments by the specified deadlines through the LMS platform, Assignments must be completed individually unless otherwise stated by the instructor. Submitted work should be clearly organized, properly formatted, and include the student's name and identification number where required. Late submissions may be subject to grade penalties or may not be accepted unless prior approval has been obtained due to documented exceptional circumstances. Students are responsible for verifying that their files have been uploaded successfully and are readable. Corrupted, incomplete, or incorrectly submitted files may be treated as non-submissions.



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	Homework solutions should demonstrate the problem-solving process, relevant calculations, algorithms, code, or explanations as appropriate. Any external sources, references, software tools, or AI-assisted contributions used in completing the assignment must be properly acknowledged in accordance with academic integrity requirements.
Late work policy	Late assignment submissions won't be accepted for grading. The grade for this assignment will be zero .
Make-up exam policy	Students who are unable to attend a quiz, midterm, or final examination due to documented medical, personal, or other exceptional circumstances must notify both the instructor and the relevant academic administration office as soon as possible. Supporting documentation must be provided for approval. Approved students may be offered a make-up examination at a time determined by the QA department and instructor. Requests submitted without valid documentation or after the examination date may not be considered. The format and coverage of the make-up examination may differ from the original assessment while evaluating the same learning outcomes.
Academic integrity expectations	Students are expected to maintain the highest standards of academic honesty and professional conduct throughout the course. All submitted assignments, laboratory reports, projects, quizzes, and examinations must represent the student's own work. Any form of plagiarism, unauthorized collaboration, cheating, fabrication of results, or misuse of artificial intelligence tools without proper acknowledgment constitutes academic misconduct and will be handled according to BHOS academic regulations. Students are responsible for correctly citing all sources of information, software libraries, datasets, and external materials used in their coursework.
Academic advising support	Students are encouraged to seek academic support whenever they encounter difficulties with course content, assignments, or study planning. Academic advising is available through the course instructor during designated office hours and by appointment. Additional guidance regarding academic progress, degree requirements, and university policies may be obtained from the Information Technology Department's Academic Adviser. Students are expected to communicate proactively with the instructor to address academic concerns and to develop effective learning strategies for success in the course.

Instructor of the course _____

Head of the department _____